

Pinned group summary: $SO(n=6, q=2)$

Pinning-Sympy automated output

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1 Basic group attributes

Group	$\mathrm{SO}(n=6, q=2)$
Matrix size	6×6
Rank	2
Defining equations	<i>DefiningEquationPlaceholder</i>

1.1 Symmetric bilinear form

Dimension	6
Witt index	2
Primitive element	N/A
Epsilon	N/A

Matrix:
$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & c_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & c_1 \end{bmatrix}$$

1.2 Generic Lie algebra element

$$\begin{bmatrix} x_{00} & x_{01} & 0 & -x_{12} & -c_0 x_{42} & -c_1 x_{52} \\ x_{10} & x_{11} & x_{12} & 0 & -c_0 x_{43} & -c_1 x_{53} \\ 0 & -x_{30} & -x_{00} & -x_{10} & -c_0 x_{40} & -c_1 x_{50} \\ x_{30} & 0 & -x_{01} & -x_{11} & -c_0 x_{41} & -c_1 x_{51} \\ x_{40} & x_{41} & x_{42} & x_{43} & 0 & x_{45} \\ x_{50} & x_{51} & x_{52} & x_{53} & -\frac{c_0 x_{45}}{c_1} & 0 \end{bmatrix}$$

1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{t_0} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{t_1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1.4 Trivial characters

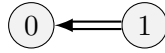
$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Summary Properties

Dynkin type	B2
Irreducible	True
Reduced	True
Simply laced	False
Number of roots	8

2.1 Dynkin diagram



2.2 Cartan matrix

$$\begin{bmatrix} 2 & -1 \\ -2 & 2 \end{bmatrix}$$

2.3 Roots and coroots

Root	Coroot	Simple	Sign	Multipliable
$(-1, -1, 0, 0, 0, 0)$	$(-1, -1, 1, 1, 0, 0)$	No	—	No
$(-1, 0, 0, 0, 0, 0)$	$(-2, 0, 2, 0, 0, 0)$	No	—	No
$(-1, 1, 0, 0, 0, 0)$	$(-1, 1, 1, -1, 0, 0)$	No	—	No
$(0, -1, 0, 0, 0, 0)$	$(0, -2, 0, 2, 0, 0)$	No	—	No
$(0, 1, 0, 0, 0, 0)$	$(0, 2, 0, -2, 0, 0)$	Yes	+	No
$(1, -1, 0, 0, 0, 0)$	$(1, -1, -1, 1, 0, 0)$	Yes	+	No
$(1, 0, 0, 0, 0, 0)$	$(2, 0, -2, 0, 0, 0)$	No	+	No
$(1, 1, 0, 0, 0, 0)$	$(1, 1, -1, -1, 0, 0)$	No	+	No

2.4 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ , the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$

α	β	Pairs (i, j)
$(-1, -1, 0, 0, 0, 0)$	$(0, 1, 0, 0, 0, 0)$	$(1, 1), (1, 2)$
$(-1, -1, 0, 0, 0, 0)$	$(1, 0, 0, 0, 0, 0)$	$(1, 1), (1, 2)$
$(-1, 0, 0, 0, 0, 0)$	$(0, -1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 0, 0, 0)$	$(0, 1, 0, 0, 0, 0)$	$(1, 1)$
$(-1, 0, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 0, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(-1, 1, 0, 0, 0, 0)$	$(0, -1, 0, 0, 0, 0)$	$(1, 1), (1, 2)$
$(-1, 1, 0, 0, 0, 0)$	$(1, 0, 0, 0, 0, 0)$	$(1, 1), (1, 2)$
$(0, -1, 0, 0, 0, 0)$	$(1, 0, 0, 0, 0, 0)$	$(1, 1)$
$(0, -1, 0, 0, 0, 0)$	$(1, 1, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(0, 1, 0, 0, 0, 0)$	$(1, -1, 0, 0, 0, 0)$	$(1, 1), (2, 1)$
$(0, 1, 0, 0, 0, 0)$	$(1, 0, 0, 0, 0, 0)$	$(1, 1)$

3 Root spaces

Root	Dimension
$(-1, -1, 0, 0, 0, 0)$	1
$(-1, 0, 0, 0, 0, 0)$	2
$(-1, 1, 0, 0, 0, 0)$	1
$(0, -1, 0, 0, 0, 0)$	2
$(0, 1, 0, 0, 0, 0)$	2
$(1, -1, 0, 0, 0, 0)$	1
$(1, 0, 0, 0, 0, 0)$	2
$(1, 1, 0, 0, 0, 0)$	1

RootSpaceTablePlaceholder

4 Root subgroups

RootSubgroupTablePlaceholder

HomomorphismDefectTablePlaceholder

HomomorphismDefectIdentitiesPlaceholder

5 Commutators

CommutatorCoefficientTablePlaceholder

CommutatorIdentitiesPlaceholder

6 Weyl group

WeylGroupPlaceholder

WeylConjugationCoefficientPlaceholder

WeylConjugationIdentitiesPlaceholder

7 Coroot torus elements (h_α)

CorootTorusElementPlaceholder